3D Matrices as Linear Transformations

Questions 1–50

Question 1. Find the image of the coordinates $(2, 3, -1)$ after the following transformations:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

f) $\begin{pmatrix} 0 & 2 & 0 \\ -3 & -5 & 4 \\ 0 & 3 & 0 \end{pmatrix}$

Question 2. A shape is transformed by the following matrices. Describe what happens in each case:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Question 3. Find the 3×3 matrix that maps $(1, 4, 2)$, $(1, 1, 0)$ and $(1, 3, 1)$ to $(11, 6, 8)$, $(3, 1, 3)$ and $(8, 4, 6)$ respectively.

Question 4. Find the 3×3 matrix that maps $(1, 1, 1)$, $(2, 1, 2)$ and $(3, 1, 1)$ to $(2, 3, 4)$, $(1, 2, 1)$ and $(0, 0, 1)$ respectively.

Question 5. It is given that the matrix
$\begin{pmatrix} 5 & 2 & -1 \\ -1 & 0 & -1 \\ 1 & 2 & 5 \end{pmatrix}$
will map (a, b, c), (d, e, f) and (g, h, i) onto $(14, 0, -16)$, $(-7, -1, 19)$ and $(-21, 1, 3)$ respectively. Find the values of a, b, c, d, e, f, g, h and i.

Question 6. It is given that the matrix $\begin{pmatrix} 2 & 1 & 2 \\ 2 & 2 & 2 \\ 1 & 2 & 1 \end{pmatrix}$ will map (a, b, c), (d, e, f) and (g, h, i) onto $(-6, -8, -6)$, $(-3, -2, 0)$ and $(6, 6, 3)$ respectively. Find the values of b, e, h, and write c, f, i in terms of a, d, g respectively.

Question 7. Describe the following linear transformations:

a) $\begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

d) $\begin{pmatrix} 7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

e) $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Question 8. A shape with volume 4 is mapped to its image by the matrix $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Write down the volume of the shape's image.

Question 9. A shape with volume 4 is mapped to its image by the matrix $\begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$. Write down the volume of the shape's image.

Question 10. The triangle T has vertices at $(10, 0, 0)$, $(0, 0, 0)$ and $(0, 6, 0)$. The matrix $\mathbf{A} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ maps T to triangle T'. Find the coordinates of the vertices and area of triangle T'.

Question 11. The triangle T has vertices at $(0, 8, 0)$, $(0, 0, 0)$ and $(0, 0, -4)$. The matrix $\mathbf{A} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ maps T to triangle T'. Find the coordinates of the vertices and area of triangle T'.

Question 12. Write down the matrix that represents each of the following linear transformations:

a) Stretch, parallel to the x-axis, scale factor 4
b) Enlargement, centred at O, scale factor 7
c) Stretch, parallel to the y-axis, scale factor $\frac{3}{5}$
d) Stretch, parallel to the x-axis, scale factor 8, and a stretch, parallel to the z-axis, scale factor 7
e) Stretch, parallel to the y-axis, scale factor $\frac{1}{4}$, and a stretch, parallel to the z-axis, scale factor $\frac{1}{8}$, and a stretch, parallel to the x-axis, scale factor $\frac{3}{4}$

Question 13. The kite K has vertices at $(0, 0, -4)$, $(-2, 0, 0)$, $(0, 0, 3)$ and $(2, 0, 0)$. K is mapped to K' by an enlargement about the origin, such that the area of K is tripled. Find the exact coordinates of the vertices of K'.

Question 14.

a) Write down the 3×3 matrix that represents a stretch, parallel to the x-axis, scale factor 4, and a stretch, parallel to the y-axis, scale factor 8, and a stretch, parallel to the z-axis, scale factor 9.

b) The point $(3m + n, n - m, m + 2n)$ is transformed by the matrix in (a) to $(16m - n, -2n, 21m + 3n)$. Given also that the sum of m and n is 9, find the values of m and n.

Question 15. Describe the following linear transformations:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

Question 16. A shape with volume 4 is mapped to its image by the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$. Write down the volume of the shape's image.

Question 17. The triangle T has vertices at $(2, 1, 1)$, $(1, 4, 1)$ and $(3, 1, 4)$. T is reflected in the plane $y = 0$ to form T'. Using matrices, find the coordinates of the vertices of T'.

Question 18. The trapezium X has vertices at $(3, 0, 0)$, $(3, 0, 5)$, $(0, 3, 0)$ and $(-2, 5, 5)$. X is reflected in the plane $x = 0$ to form X'. Using matrices, find the coordinates of the vertices of X'.

Question 19. Write down the 3×3 matrix that represents each of the following linear transformations:

a) A reflection in the plane $z = 0$
b) A reflection in the plane $y = 0$
c) A reflection in the plane $x = 0$
d) A reflection in the plane $x = y$
e) A reflection in the plane $x = z$
f) A reflection in the plane $y = z$

Question 20. Given $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 - 2k^2(2k+3) \end{pmatrix}$ represents a reflection in the plane $z = 0$, for constant k, find the possible values of k.

Question 21. Describe the following linear transformations:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

d) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$

f) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$

g) $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

h) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

i) $\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Question 22. A shape with volume 4 is mapped to its image by the matrix $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$. Write down the volume of the shape's image.

Question 23. The point $A(7, -2, 1)$ is rotated by $90°$ anticlockwise about the z-axis to point B. Find the coordinates of point B.

Question 24. The point A is rotated by $180°$ anticlockwise about the z-axis to point $B(3, -4, 5)$. Find the coordinates of point A.

Question 25. The triangle T has vertices at $(-1, 1, 3)$, $(2, 5, 1)$ and $(4, -3, -2)$. T is rotated by $180°$ about the y-axis to form T'. Find the coordinates of the vertices of T'.

Question 26. The point P is rotated by $270°$ anticlockwise about the x-axis to point $Q(a, b, c)$, where a, b, c are positive constants. Find the coordinates of point P in terms of a, b and c.

Question 27. Write down the matrix that represents the following linear transformations:

a) A rotation by 45°, anticlockwise about the x-axis
b) A rotation by 135°, anticlockwise about the y-axis
c) A rotation by 315°, anticlockwise about the z-axis

Question 28. Write down the matrix that represents the following linear transformations:

a) A rotation by 30°, anticlockwise about the x-axis
b) A rotation by 120°, anticlockwise about the y-axis
c) A rotation by 150°, anticlockwise about the z-axis
d) A rotation by 60°, clockwise about the x-axis
e) A rotation by 225°, clockwise about the y-axis
f) A rotation by 330°, clockwise about the z-axis

Question 29. Write down the matrix that represents a rotation by 35°, anticlockwise about the y-axis. Give each of its elements to 3 significant figures.

Question 30. Write down the matrix that represents a rotation by 102°, clockwise about the x-axis. Give each of its elements to 3 significant figures.

Question 31. The triangle T has vertices at $(3, 0, 9)$, $(-2, 1, 0)$ and $(0, 5, -4)$. T is rotated by $45°$ anticlockwise about the z-axis to form T'. Find the exact coordinates of the vertices of T'.

Question 32. The rectangle R has vertices at $(0, 3, -1)$, $(2, 0, -1)$, $(2, 0, 2)$ and $(0, 3, 2)$. R is rotated by $135°$ clockwise about the x-axis to form R'. Find the exact coordinates of the vertices of R'.

Question 33. The point $A(16, -64, 4)$ is rotated anticlockwise about the z-axis by an angle θ, where θ is an obtuse angle, so that its image A' has x-coordinate 0. Find the exact coordinates of A'.

Question 34. The triangle X has vertices at $(1, 0, k\sqrt{2})$, $(0, \frac{\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2})$ and $(2, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$, where k is a positive constant. X is rotated by 45° clockwise about the x-axis to form X', which has vertices $(1, 2, 2)$, $(0, -1, -2)$ and $(2, 0, -1)$ respectively. Find the value of k.

Question 35. Each of the following matrices represents a rotation. Describe fully the single transformation given by:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

b) $\begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$

c) $\begin{pmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{pmatrix}$

d) $\begin{pmatrix} \frac{\sqrt{6}+\sqrt{2}}{4} & 0 & \frac{\sqrt{2}-\sqrt{6}}{4} \\ 0 & 1 & 0 \\ \frac{\sqrt{6}-\sqrt{2}}{4} & 0 & \frac{\sqrt{6}+\sqrt{2}}{4} \end{pmatrix}$

e) $\begin{pmatrix} \frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{2}-\sqrt{6}}{4} & 0 \\ \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{6}+\sqrt{2}}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Question 36. The matrix $\begin{pmatrix} \frac{1}{3} & 0 & \frac{2\sqrt{2}}{3} \\ 0 & 1 & 0 \\ -\frac{2\sqrt{2}}{3} & 0 & \frac{1}{3} \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the y-axis. Find the value of θ to 3 significant figures.

Question 37. The matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{4} & -\frac{3}{4} \\ 0 & \frac{3}{4} & \frac{\sqrt{7}}{4} \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the x-axis. Find the value of θ to 3 significant figures.

Question 38. Given $\mathbf{A} = \begin{pmatrix} \frac{\sqrt{6}-\sqrt{2}}{4} & -\frac{\sqrt{6}+\sqrt{2}}{4} & 0 \\ \frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{6}-\sqrt{2}}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the z-axis, find the smallest integer n such that $\mathbf{A}^n = \mathbf{I}$.

Question 39. The point $A(-3, 4, 7)$ is rotated by $\theta°$ anticlockwise about the x-axis to the point $B(-3, -7, -4)$.
Find the value of θ to 3 significant figures.

Question 40. The point $A(1, 8, -6)$ is rotated by $\theta°$ anticlockwise about the y-axis to the point $B(-6, 8, 1)$. Find the value of θ to 3 significant figures.

Question 41. The point $A(4, 8, 2)$ is rotated by $\theta°$ anticlockwise about the z-axis to the point $B(-8, -4, 2)$. Find the value of θ to 3 significant figures.

Question 42. The point $A(5, -5, -6)$ is rotated by $\theta°$ anticlockwise about an axis to the point $B(6, -5, -5)$. Find the value of θ to 3 significant figures.

Question 43. It is given that the matrices $\mathbf{A} = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix}$ represent the linear transformations A and B respectively. Find the matrix that represents the combined transformation:

a) A followed by B

b) B followed by A

Question 44. Find the matrix that represents a reflection in the plane $x = 0$ followed by a reflection in the plane $z = 0$. Hence describe the single transformation represented by this matrix.

Question 45. It is given that the matrix $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$

a) Describe fully the single transformation represented by $\mathbf{A}$.

b) Describe fully the single transformation represented by $\mathbf{A}^{2025}$.

Question 46. It is given that the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

a) Describe fully the single transformation represented by $\mathbf{A}$.

b) Describe fully the single transformation represented by $\mathbf{A}^{443}$.

Question 47. Determine the matrix that represents:

a) An enlargement, centred at the origin, by scale factor 3, followed by a stretch, parallel to the z-axis, scale factor 5

b) A reflection in the plane $y = 0$, followed by a reflection in the plane $x = 0$

c) A rotation by 90° anticlockwise about the x-axis, followed by a stretch, parallel to the y-axis, scale factor 5

d) A reflection in the plane $z = 0$, followed by a rotation by 180° about the y-axis

e) A reflection in the plane $y = 0$, followed by a rotation by 90° clockwise about the z-axis, followed by a reflection in the plane $x = 0$

Question 48. Given $\mathbf{A} = \begin{pmatrix} -k & m & 0 \\ \frac{4k}{m} & k & 0 \\ 0 & 0 & -k\sqrt{3} \end{pmatrix}$ and that $\mathbf{A}^2$ represents an enlargement, centred at the origin, by scale factor 12, find the value of k.

Question 49. It is given that $\mathbf{A} = \begin{pmatrix} 6 & 3 & 4 \\ -2 & 2 & 4 \\ 5 & 4 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$. It is also given that $\mathbf{AB}$ represents a rotation by 180° about the z-axis, followed by an enlargement, centred at the origin, by scale factor 5. Find the values of a, b, c, d, e, f, g, h and i.

Question 50. The matrix $\mathbf{A}$ represents a composition of two linear transformations. One of the transformations is an enlargement, centred at the origin, by scale factor k, where k is a positive constant. Fully describe both transformations in each of these situations:

a) $\mathbf{A} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 4 \end{pmatrix}$

b) $\mathbf{A} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

c) $\mathbf{A} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

d) $\mathbf{A} = \begin{pmatrix} 0 & 8 & 0 \\ -8 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$